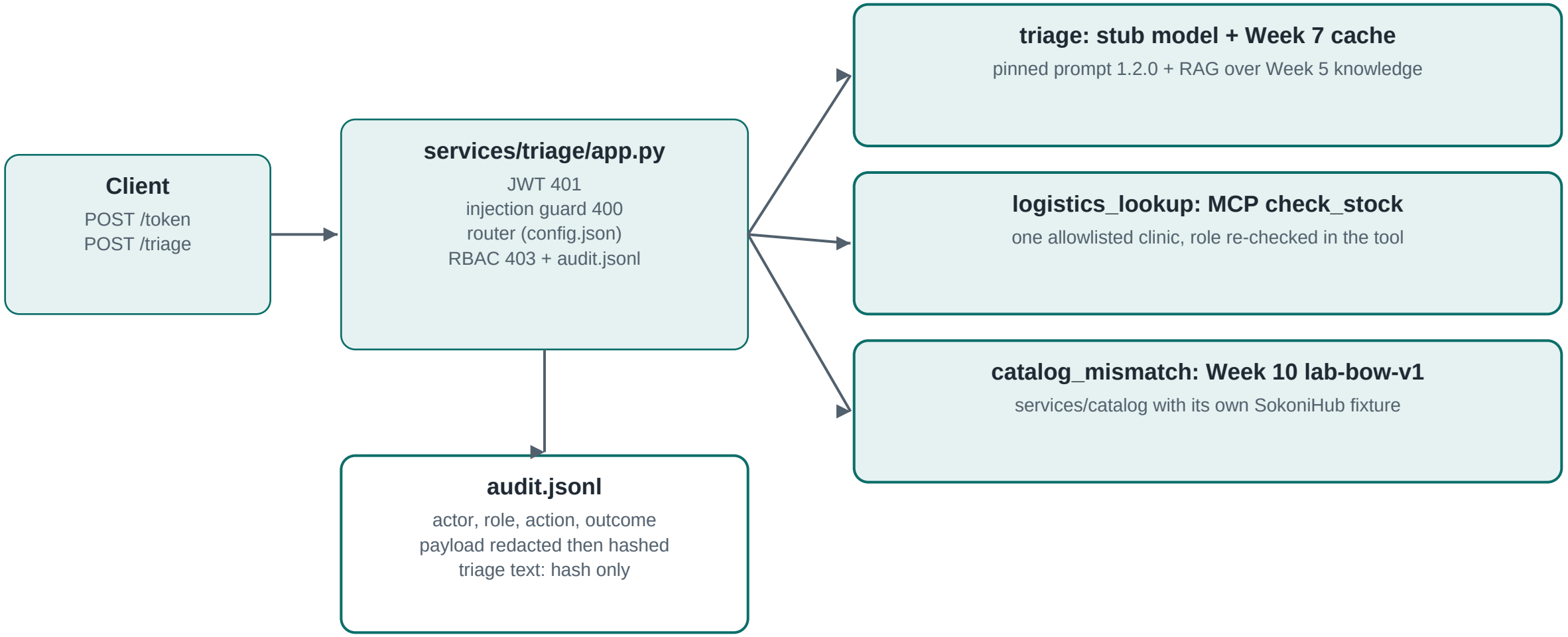
One door for clinic coordinators

- Clinic coordinators juggle three separate tools from this module: triage, stock and listing checks
  - each with its own login, rules and logs
- User: AfyaPlus clinical_ops staff and partner clinics in western Kenya
- Decision supported: what happens next (send to clinic, reorder, review a listing)
- The system recommends; people decide. No diagnosis, no prescribing

Official AfyaPlus path, real artefacts

- AfyaPlus Module 4 Capstone (not the domain alternative)
- Reuse, not rewrite: Week 5 knowledge, Week 6 API + JWT, Week 7 MCP tool + cache,
  - Week 8 pinned prompt (same sha 85e5cb1d...), Week 9 security, Week 10 catalog scorer
- Stub mode everywhere: deterministic tests, no API key, $0 this week
- Dataset: 300 synthetic intent rows + 10 SokoniHub listings in a separate folder

Architecture and module boundaries



Data boundary: fixtures/ (clinical, synthetic) and services/catalog/fixtures/ (marketplace) never mix. tests/unit/test_fixture_separation.py fails the build if they do.

Skill coverage: every skill has a file

Skill	Week	Where in the repo
API or agent path	6	services/triage/app.py, auth.py
Cost / MCP tools	7	agents/tools/logistics.py, model.py cache
Pin / versioning	8	services/triage/prompts/pin.json, corpus_pin.json
Security hygiene	9	tests/test_security_smoke.py, services/triage/security/
RAG / grounded lookup	5	services/rag/retriever.py + knowledge/
Eval / test	11	tests/unit, integration, perf; evals/golden.jsonl
Optional multimodal	10	services/catalog (own fixtures)

integration_map.json: 10 artefacts, every path checked on disk by check_integration_map.py

Live demo: python run_demo.py

Request	Result
GET /health	200, prompt 1.2.0 sha 85e5cb1d..., model triage-stub-v1
"My child has had a fever since yesterday"	medium, grounded in faq-fever
"Chest pain and difficulty breathing"	high (red-flag floor), faq-danger-signs
"How much ORS stock is at C04?"	MCP check_stock: 5 units, reorder needed
"stock of ORS at *"	refused: one allowlisted clinic only (400)
"Check the listing photo for NK-99"	Week 10 lane: flagged for review
no token / partner triage / injection	401 / 403 / 400

Testing evidence

- Unit (default): 45 passed - router, gold fixture, service functions, MCP tool, pins
- Security smoke: 9 passed - redaction, JWT 401, injection 400, RBAC 403, clean audit
- Integration -m integration: 4 passed + live httpx /health; perf -m perf: p95 0.025 s
- Golden-set eval: 30/32 = 0.94, gate 0.90; both misses fall back to triage
- Fixture incident: test went red on 28 rows, router fixed (not the test), green again

Security and compliance choices

- Authentication is not authorisation: JWT on /triage, then a role matrix (403)
- Refuse before spending: instruction-like text gets 400 before any tool or model
- No bulk export: check_stock answers for one allowlisted clinic; "*" refused
- Audit log redacted then hashed; health text kept as a hash only
- Pins on the prompt and the knowledge corpus; edits fail closed at start-up
- No secrets in git history (tested); clinical and marketplace fixtures never mixed

Cost table with assumptions

Call type	Calls	Total	Assumption
Tests, eval, demo (stub)	0 live	$0	no key used this week
Live demo, if wired	~10	~$0.0004	$0.0000371 per call (Week 7 tokens)
2,000 triage/day	60k/mo	~$2.23/mo	gpt-4o-mini list price
+ Week 7 cache (0.60 hits)	24k/mo	~$0.89/mo	hit rate from Week 7 fixture

Tokens are about 1% of the all-in cost (Week 7): people and hosting dominate.

What works vs what is deferred

Works now

- One command, end to end, stub mode
- Auth, RAG, MCP tool, catalog lane
- Pins, smoke, eval in CI
- Measured perf, fixture checks

Deferred (said, not claimed)

- Live LLM (no key; stub only)
- Cloud / Docker deploy (not executed live)
- LLM agent loop over MCP
- Week 4 fine-tuned model
- Classmate peer review (solo, declared)

Process and next steps

- Three feature branches -> PR #1 Monday, PR #2 Tuesday, PR #3 Thursday
- Each PR: template, CI green, checklist review comment, merge commit
- review_log.md: date, PR, reviewer (solo, declared), outcomes and follow-ups
- Next: a classifier router gated by the golden set
- Then: the live model behind a flag, with the stub kept for CI